

The Audit Said the Model Was Fine

It was auditing 2,196 people. There were 10,978.

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1 WHY WE DID THIS

We set out to answer a practical question: *can a language model be trusted to interpret fairness metrics?* Companies are already wiring LLMs into compliance tooling, and nobody had built ground truth that wasn't itself a model's opinion. So we built it — a deterministic pipeline that computes every fairness number and assigns every severity label by fixed rule, with the model allowed to interpret and nothing else.

Then we noticed we were validating a model against an audit pipeline that nobody had validated. So we audited the audit. That is this paper.

2 THE RESULT THAT REFRAMED THE PROJECT

Our mortgage audit ranked **American Indian men as the best-treated group** — disparate impact 1.15, above parity, nothing to see. It was looking at four of them. All four happened to be approved.

Auditing all 10,978 applicants instead of a 2,196-row sample, the same model on the same data puts that group at 0.6734: **the worst group in the dataset**, in the CRITICAL band, 31 people.

The audit did not merely miss a disparity. It **inverted** it, and reported the inversion with a clean severity label. Nothing was broken — the metrics were right, the code was right, the reporting floor that hid the small groups was statistically correct to hide them. The sample was simply too small for the affected groups to be visible at all, and nothing in the output said so.

3 WHAT ELSE THE AUDIT'S OWN DEFAULTS DECIDED

4 WHY THIS MATTERS

Audits are becoming gates. Bias-audit requirements are entering regulation, procurement, and model sign-off. Increasingly a system ships because an audit says so. Every failure above produced a *clean* result — the direction nobody double-checks. A false alarm costs a reviewer an afternoon; a false clearance closes the question permanently.

Nobody audits the auditor. These were not bugs and nobody was careless. They are defaults — chosen once, applied everywhere, visible only in source code. When an audit was a human report, its assumptions were arguable. When an audit is a pipeline, its assumptions are invisible.

This is not just fairness. Our headline model score was 75%; it decomposes to 33%/92%/100%, where the 100% is the case so easy that agreeing costs nothing. Any benchmark averaging over uneven tasks can report difficulty as if it were capability.

Table 1: Each is a reasonable engineering decision. Each silently answers a normative question.

Default	What it decided
Which age counts as protected	The pipeline used 40+ — that is <i>employment</i> law. Credit law (ECOA) protects 62+. On the right cut the model shows the largest age disparity in the data (0.8897); on the wrong one it looks clean (0.9791).
Which outcome is "good"	Computed on the adverse label of a column named <code>loan_default</code> , disparate impact read 0.0 — maximum alarm. The true value: 0.9931, near parity.
What counts as an answer	A 40-character minimum marked 6 correct answers as refusals. It then zeroed 14 evaluations of a second AI provider, dropping its score 70.0 → 19.79 — and we permanently blocked that vendor on the strength of it.
What counts as evidence	A fabrication check flagged the newer model 5× more often. 67% of those flags were the model doing arithmetic it was asked to do (88.88% is just 0.8888). It penalised showing your work.

5 HOW TO USE THIS

If you run audits: audit your whole population, not a convenience sample. Print what your size floor *withheld*, with n — ours printed only what passed. Put the statute citation next to the threshold in code. State the favorable outcome explicitly; never infer it from label values.

If you build audit tooling: treat every mechanical gate as a hypothesis and test it on the cases where the right answer is unusual — the terse answer, the six-person subgroup, the model that shows its work. All three of ours failed exactly there.

If you buy or regulate audits: ask what sample the audit ran on and what it suppressed. An audit that cannot answer those two questions cannot support the clearance it is being used to give.

If you want to reuse it: the pipeline, the frozen prompts and every number here are released and regenerate offline, no API key, for \$0.89 of compute.

6 ARTIFACTS

Full results artifacts/consolidated/RESULTS.md · paper report/report_facct.tex · sample-size analysis scripts/compare_populations.py. 60 tests, CI green.